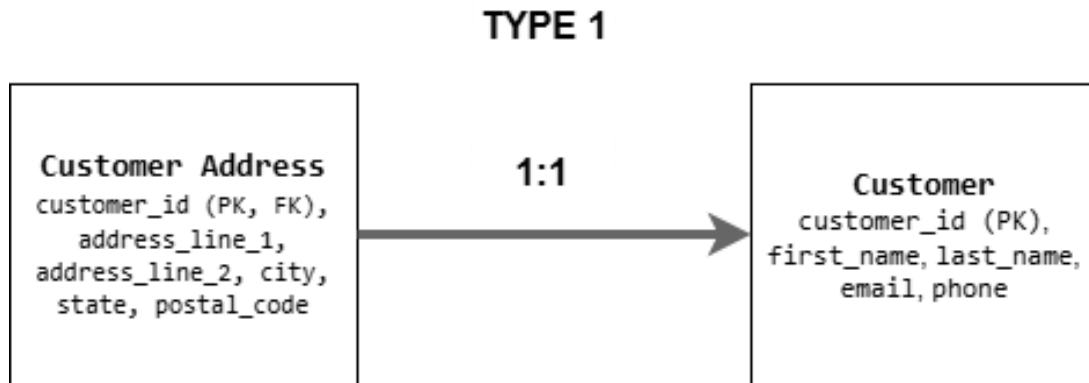


Prompt 3

Architecture 1: Overwriting Changes

This architecture always keeps the most recent address for a customer. Previous addresses are overwritten when a new address is added.

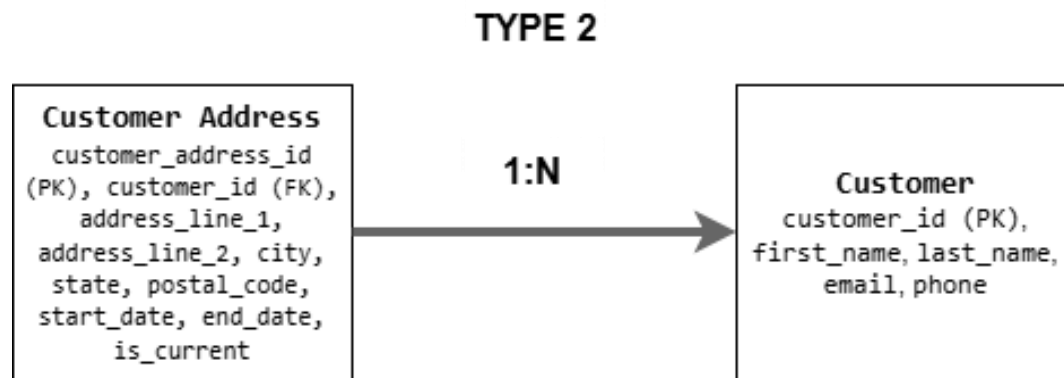


SQL Code:

```
CREATE TABLE customer_address (  
    customer_id INTEGER PRIMARY KEY REFERENCES customer(customer_id),  
    address_line_1 TEXT NOT NULL,  
    address_line_2 TEXT,  
    city TEXT NOT NULL,  
    state TEXT NOT NULL,  
    postal_code TEXT NOT NULL  
);
```

Architecture 2: Retaining Changes

This architecture tracks historical addresses for each customer by creating a new row for every address change.



SQL Code:

```
CREATE TABLE customer_address (  
    customer_address_id INTEGER PRIMARY KEY,  
    customer_id INTEGER NOT NULL REFERENCES customer(customer_id),  
    address_line_1 TEXT NOT NULL,  
    address_line_2 TEXT,  
    city TEXT NOT NULL,  
    state TEXT NOT NULL,  
    postal_code TEXT NOT NULL,  
    start_date DATE NOT NULL,  
    end_date DATE,  
    is_current BOOLEAN DEFAULT TRUE  
);
```

Comparison of Type 1 vs. Type 2

Feature	Type 1 (Overwrite)	Type 2 (Retain Changes)
Definition	Overwrites existing data	Maintains historical records
Complexity	Simple	More Complex
Storage	Minimum	Larger
Use Case	Latest/current data only	Historical tracking needed